Main Model Estimation & Robustness

ECON 692 – Applied Economics Seminar

Andrew Hobbs

University of San Francisco

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Today's Agenda

Shortened class today — we end at 6:00. Tech Economics Seminar reception at 6:30.

Time	Duration	Activity
4:35	10 min	Check-in: where are you on results?
4:45	15 min	Estimation & robustness quick hits
5:00	45 min	Work sprint: finish your estimation
5:45	15 min	Checkpoint, wrap-up & next steps

Preliminary Results due tomorrow (Friday 3/27).

Check-in

Where are you on your preliminary results?

Quick round: one sentence on your status.

- Have you run your main specification?
- Can you interpret the coefficient?
- What is blocking you?

Today's Goal

Leave class with a **working main specification** and at least one **robustness check** — ready to write up tomorrow.

Estimation & Robustness Quick Hits

Learning Objectives

1. **Run** your main specification and get a coefficient
2. **Interpret** the coefficient in one sentence
3. **Identify** at least one robustness check for your design
4. **Submit** preliminary results that meet the minimum standard

What “Preliminary Results” Should Look Like

Minimum deliverable:

- Main specification estimated
- Coefficient interpreted in context
- At least one robustness check
- Code is reproducible (`git push`)

Stretch goals:

- Event-study plot (if applicable)
- Multiple specs showing coefficient stability
- Draft results section (2–3 paragraphs)
- Formatted table of results

Key

This is not a final paper. It is proof that your **empirical strategy works on your data** and produces interpretable results.

Common Estimation Pitfalls

1. **Wrong standard errors** — Panel data ⇒ cluster at the treatment level. `feols(..., cluster = ~state)` not `vcov = "HC1"`
2. **Fixed effects absorb your treatment** — If treatment only varies by year, year FE kill it. You need **cross-sectional variation.**
3. **Collinearity warnings** — A variable is a linear combination of your FE. Drop it or rethink the spec.
4. **"No variation in treatment"** — Check: is the treatment variable coded correctly? Are the right units treated?
5. **Sample size drops dramatically** — Merges or missing values silently dropping observations. Check `nobs()`.

Robustness: What Counts

Your Method	Core Robustness Check	Bonus
DiD / Event Study	Pre-trends test; placebo treatment date	Alt. control group; drop never-treated
Synthetic Control / SDID	Pre-period fit; placebo units	Leave-one-out donors
Panel FE	Add controls sequentially; coefficient stability	Oster bounds
IV / 2SLS	First-stage F > 10; reduced form	Placebo instrument
BSTS	Placebo treatment dates; covariate balance	Vary model priors

Pick one from the "Core" column. That's your minimum for tomorrow.

Work Sprint

Work Sprint: Finish Your Estimation

45 minutes — hands on

Setup (first 5 minutes):

1. Open your project in the terminal
2. Ensure `CLAUDE.md` includes your empirical strategy
3. Pull your latest code: `git pull`

Tasks (in order):

1. Run your **main specification** — get a coefficient
2. Write **one sentence** interpreting that coefficient
3. Run one **robustness check** from the table on the previous slide
4. `git add`, `git commit`, `git push`

Raise your hand if stuck. I will circulate.

Sprint: Method-Specific Starting Points

Your Method	First Command	Then Ask Claude
DiD / Event Study	`feols(y ~ treat:post | id + t)`	"Add an event study"
SDID	`synthdid_estimate(Y, N0, T0)`	"Run placebo tests"
BSTS	`CausalImpact(data, pre, post)`	"Plot the counterfactual"
Panel FE	`feols(y ~ x | id + t)`	"Add controls one at a time"
IV / 2SLS	`feols(y ~ 1 | id + t | x ~ z)`	"Report first-stage F"

If you get stuck on data issues, ask Claude: "My data has [problem]. How do I fix it before running the regression?"

Sprint Checkpoint

Before you leave, verify:

- ☐ I ran my main specification and got a coefficient
- ☐ I can state in one sentence what the coefficient means
- ☐ I ran at least one robustness check
- ☐ My code is saved and pushed (`git commit` + `git push`)

Checked all four? You are in good shape for **Preliminary Results** tomorrow.

Missing something? You have tonight to finish. Office hours are open — email me.

Wrap-Up & Next Steps

Preliminary Results: Due Tomorrow

Due: Friday, March 27 at 11:59pm

Submit:

1. Your **main regression output** (table or screenshot)
2. A **one-paragraph interpretation** of your main result
3. At least one **robustness check** with output
4. Your **code** (pushed to GitHub, reproducible)

This does not need to be polished. It needs to **work.**

Tech Economics Seminar Reception

Tonight at 6:30

Great opportunity to connect with speakers and fellow economists.

I encourage you all to attend.

Looking Ahead

Week	Date	Focus
8	**Mar 26**	**Main model estimation; robustness (today)**
9	Apr 2	Interpreting results; crafting the empirical narrative
10	Apr 9	Writing: intro, background, empirical strategy
11	Apr 16	Figures, tables, slides; presentation prep

Next week: You have results. Now we learn how to *interpret* and *present* them.

Wrap-Up

1. **Preliminary results** are proof your strategy works on your data
2. Pick the **right robustness check** for your method — one is enough for now
3. **Preliminary Results due tomorrow** (Friday 3/27)
4. Next week: from results to **narrative**

Good luck finishing up tonight. See you next Thursday.